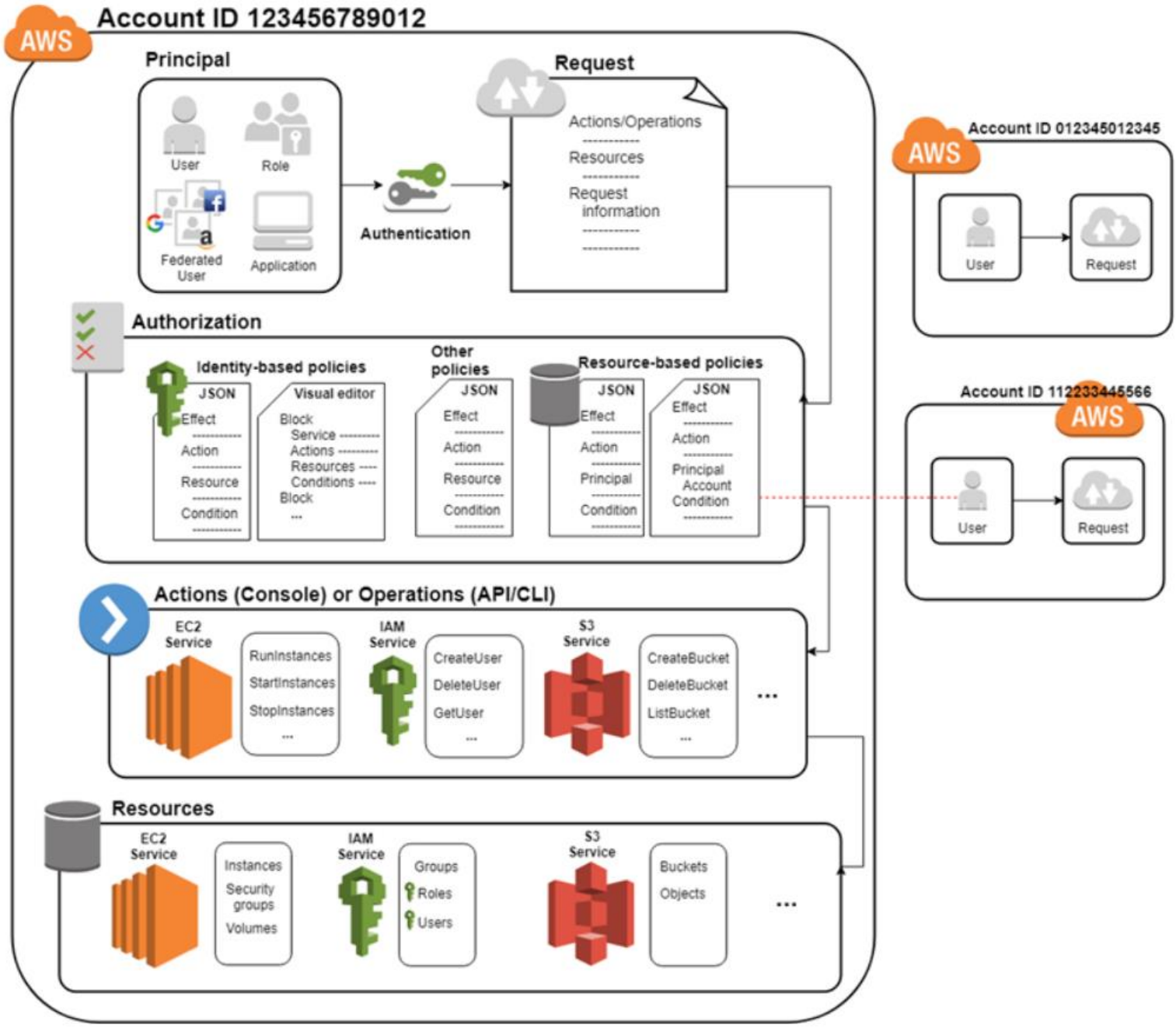
How IAM works

IAM provides the infrastructure necessary to control authentication and authorization for your AWS account. The IAM infrastructure is illustrated by the following diagram:

First, a human user or an application uses their sign-in credentials to authenticate with AWS. Authentication is provided by matching the sign-in credentials to a principal (an IAM user, federated user, IAM role, or application) trusted by the AWS account. Next, a request is made to grant the principal access to resources. Access is granted in response to an authorization request. For example, when you first sign in to the console and are on the console Home page, you are not accessing a specific service. When you select a service, the request for authorization is sent to that service and it looks to see if your identity is on the list of authorized users, what policies are being enforced to control the level of access granted, and any other policies that might be in effect. Authorization requests can be made by principals within your AWS account or from another AWS account that you trust.



Principals

A person or application that uses the AWS account root user, an IAM user, or an IAM role to sign in and make requests to AWS. Principals include federated users and assumed roles.

Human users

Also known as *human identities*; the people, administrators, developers, operators, and consumers of your applications.

Workload

A collection of resources and code that delivers business value, such as an application or backend process. Can include applications, operational tools, and components.

A *principal* is a human user or workload that can make a request for an action or operation on an AWS resource. After authentication, the principal can be granted either permanent or temporary credentials to make requests to AWS, depending on the principal type. IAM users and root user are granted permanent credentials, while roles are granted temporary credentials. As a best practice, we recommend that you require human users and workloads to access AWS resources using temporary credentials.

Request

When a principal tries to use the AWS Management Console, the AWS API, or the AWS CLI, that principal sends a *request* to AWS. The request includes the following information:

- **Actions or operations** – The actions or operations that the principal wants to perform. This can be an action in the AWS Management Console, or an operation in the AWS CLI or AWS API.
- **Resources** – The AWS resource object upon which the actions or operations are performed.
- **Principal** – The person or application that used an entity (user or role) to send the request. Information about the principal includes the policies that are associated with the entity that the principal used to sign in.
- **Environment data** – Information about the IP address, user agent, SSL enabled status, or the time of day.
- **Resource data** – Data related to the resource that is being requested. This can include information such as a DynamoDB table name or a tag on an Amazon EC2 instance.

AWS gathers the request information into a *request context*, which is used to evaluate and authorize the request.

Authentication

A principal must be authenticated (signed in to AWS) using their credentials to send a request to AWS. Some services, such as Amazon S3 and AWS STS, allow a few requests from anonymous users. However, they are the exception to the rule.

Once authorized, the principal can take action or perform operations on resources in your AWS account. For example, the principal could launch a new Amazon Elastic Compute Cloud instance, modify IAM group membership, or delete Amazon Simple Storage Service buckets.

Terms

These IAM terms are commonly used when working with AWS:

IAM Resource

IAM resources are stored in IAM. You can add, edit, and remove them from IAM.

- user
- group
- role
- policy
- identity-provider object

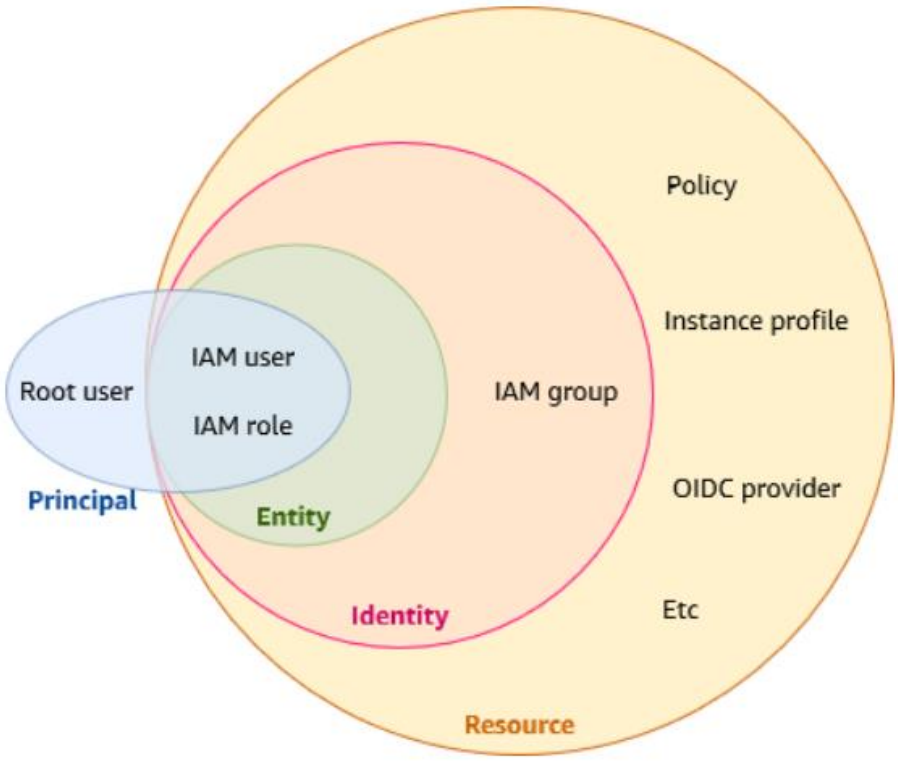
IAM Entity

IAM resources that AWS uses for authentication. Entities can be specified as a Principal in a resource-based policy.

- user
- role

IAM Identity

An IAM resource that can be authorized in policies to perform actions and to access resources. Identities include users, groups, and roles.



To authenticate from the console as a root user, you must sign in with your email address and password. As a federated user, you are authenticated by your identity provider and granted access to AWS resources by assuming IAM roles. As an IAM user, provide your account ID or alias, and then your user name and password. To authenticate workloads from the API or AWS CLI, you might use temporary credentials through being assigned a role or you might use long-term credentials by providing your access key and secret key. You might also be required to provide additional security information. As a best practice, AWS recommends that you use multi-factor authentication (MFA) and temporary credentials to increase the security of your account. To learn more about the IAM entities that AWS can authenticate, see IAM users and IAM roles.

Authorization

You must also be authorized (allowed) to complete your request. During authorization, AWS uses values from the request context to check for policies that apply to the request. It then uses the policies to determine whether to allow or deny the request. Most policies are stored in AWS as JSON documents and specify the permissions for principal entities. There are several types of policies that can affect whether a request is authorized. To provide your users with permissions to access the AWS resources in their own account, you need only identity-based policies. Resource-based policies are popular for granting cross-account access. The other policy types are advanced features and should be used carefully.

AWS checks each policy that applies to the context of your request. If a single permissions policy includes a denied action, AWS denies the entire request and stops evaluating. This is called an *explicit deny*. Because requests are *denied by default*, AWS authorizes your request only if every part of your request is allowed by the applicable permissions policies. The evaluation logic for a request within a single account follows these general rules:

- By default, all requests are denied. (In general, requests made using the AWS account root user credentials for resources in the account are always allowed.)
- An explicit allow in any permissions policy (identity-based or resource-based) overrides this default.
- The existence of an Organizations SCP, IAM permissions boundary, or a session policy overrides the allow. If one or more of these policy types exists, they must all allow the request. Otherwise, it is implicitly denied.
- An explicit deny in any policy overrides any allows.

After your request has been authenticated and authorized, AWS approves the actions or operations in your request. Operations are defined by a service, and include things that you can do to a resource, such as viewing, creating, editing, and deleting that resource. For example, IAM supports approximately 40 actions for a user resource, including the following actions: `Create User`, `Delete User`, `Get User`, `Updates` allow a principal to perform an operation, you must include the necessary actions in a policy that applies to the principal or the affected resource. To see a list of actions, resource types, and condition keys supported by each service, see Actions, Resources, and Condition Keys for AWS Services.